



Silicon recycling breakthrough

New tech recovers pure silicon from end-of-life solar cells.

<https://digit.in/dec21-15>



AI Heisenberg

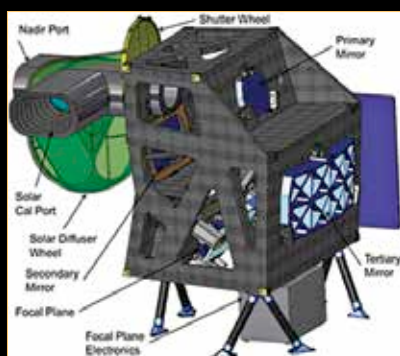
Researchers have trained a computer to predict the molecular structure of new drugs.

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nies. The funding for Landsat 9's first development also includes research into lower-cost, smaller alternatives for the future of Landsat gear.

Overall, the construction of the aircraft is based on the company's Low Earth Orbit spacecraft LEO-Star-3, which has been successfully tested in NASA's Fermi and Swift Gamma-ray observatories. On the body of Landsat 9, there are two large sensors: the Operational Land Imager-2 (OLI-2) built by Ball Aerospace and Technologies Corporation, and the Thermal Infrared Sensor-2 (TIRS-2) built at NASA's Goddard Space Flight Center.

The OLI-2: The Operational Land Imager-2 collects reflective band data from the Earth. It has five main components: the Shutter wheel assembly, earth-view baffle, solar-view baffle, diffuser assembly and focal plane electronics. The earth-view baffle element is used to view the Earth. Light enters the telescope via a hole in the shutter wheel during nominal observations. In total, two lamp assemblies with six small lamps each illuminate the full focal plane of the telescope with the shutter closed. Additionally, the



solar-view baffle points towards the sun so that the diffuser panel can reflect illumination into the telescope for the comparison and calibration of data. Functionally, it is a photosensitive detector with a 15-degree field-of-view covering a 185 km (115 mi) across-ground area. In Landsat 9, the photosensitive detectors are divided into 14

modules. Silicon PIN (SiPIN) is used to make the visible and near-infrared detectors. Mercury–Cadmium–Telluride (MgCdTe) is used to make the shortwave infrared detectors. In the OLI-2, for each spectral band there is a specific filter. All these filters are arranged together like a “butcher-block” over each module's detector range. Landsat 9's OLI-2 has a significantly improved signal-to-noise ratio over Landsat 8's OLI.

The TIRS-2: The Thermal Infrared Sensor-2 collects thermal infrared bands as emissions from the Earth's surface. It is a Class-B instrument with a five-year design life to Landsat 8's TIRS having a three-year design life. All-in-all, TIRS-2 has six main components: The Earth view, the Earth shield, black body radiator, cryocooler radiator and focal plane electronics. The focal plane has long arrays of photosensitive detectors.

A four-element refractive telescope concentrates an f/1.64 beam of thermal radiation onto a cryogenically cooled focal plane. Just like OLI-2, the TIRS-2 has a 185 km across-track swath and 15-degree field-of-view. Three modules with quantum-well infrared photodetector (QWIP) arrays are positioned in an alternating pattern along the focal plane centerline in the TIRS-2 focal plane. Using a mirror operated by a scene choice mechanism, the field-of-view is flipped between the Earth. Both an internal blackbody and a far space image is used for on-orbit radiometric calibration. This permits the view to be adjusted without affecting the Landsat 9 spacecraft's nominal earth-viewing attitude.

The instruments in Landsat 9 will take around 700 new photos every day at a resolution of 30 metres across 11 wavelengths. This resolution is fine enough to distinguish small agricultural fields, houses, forest stands, and lakes while still covering enormous regions of terrain. The devices' 11 wavelengths will record data about Earth in the visible to thermal infrared ranges, including

data on coastal waters, forests, and wildfire scars.

THE LANDSAT DATASET

Landsat 8 has collected data for 3.5 years. In this process, it has already added over 8,27,000 images to the dataset, which is 12.5 percent of data collected over 44 years. Similar to this, Landsat 9 will add radiometrically and geometrically better data than previous generation Landsats. Together, these satellites will provide new data every 8 days.

There are multiple ways to view and process Landsat data. The data can be searched and downloaded from USGS EarthExplorer, Glovis, LandsatLook Viewer and more.

LANDSAT COLLECTION

The USGS redesigned the Landsat archive in 2016 into a collection management system called Landsat Collection 1. This structure guarantees that all Landsat Level-1 products provide a consistent archive of known data quality, while also allowing for continual archive improvement and access to all data as it is acquired. Collection 1 covers all Level-1 data acquired from Landsat 1-8 from 1972 to the present.

Landsat Collection 2 is the USGS Landsat Level-1 archive's second major reprocessing event, resulting in various data product upgrades that make use of recent advances in data processing, algorithm development, and data access and dissemination capabilities.

Landsat's capacity to grasp how the Earth's land surface changes through time is one of its most valuable contributions. This data provides insight into things like glacier retreat in Alaska, crop health in California, and volcano hot areas in Yellowstone in ways never before imaginable. Its photographs can even be found on Google Earth, where millions of people can view them from their computers and smartphones. And so Landsat looked, and the world saw. **d**